Revisiting semiclassical scalar QED in 1+1 dimensions

Authors

Abstract

In this article we study the backreaction of a charged quantum scalar field on a background classical constant electric field in a finite one-dimensional region, with attention to boundary effects. We review previous work in this setup [1], where it was found that vacuum polarization averts instabilities due to runaway solutions. Their prescription for computing the vacuum polarization was later pointed out to be incorrect [11], and a correct, gauge-invariant definition using Hadamard point-splitting was introduced [15], yielding a vacuum polarization an order of magnitude weaker. Applying this correct prescription through an iterative procedure across varying boundary conditions and masses, we find that the system remains stable despite the weaker vacuum polarization. Interestingly, the numerical discrepancies between the two prescriptions do not extend to the strong-field regime, where their behavior is remarkably similar; we attribute this to the dominant low-energy modes being the most sensitive to the background field.

Contents

1	Introduction	1
2	Set up	2
2.1	Field modes	2
2.2	External Field Approximation and Critical Field Strength	4
2.3	Mode sum formula	4
3	Self-consistent fields	5
4	Results	6
4.1	Mode sum formula	8
5	Discussion and outlook	8
A	Convergence of the self-consistent fields	9

1 Introduction

Vacuum polarization is a phenomenon in quantum electrodynamics (QED) by which the vacuum of a charged field polarizes when subject to external electric fields, effectively behaving like a dielectric with $\epsilon > 0$. This mechanism was one of the first QED effects studied [3], with consequences that followed closely [14; 6]. This mechanism has been already observed, e.g. in corrections to atomic levels [9] (cf. Ref. [10] for a review), while other effects e.g. the Schwinger effect [12] lay beyond current technological advances (cf. Ref. [4]).

Defining vacuum polarization for generic backgrounds[1] is no easy task. Historically, [3] first defined it through the assumption that the short-distance behavior of the physically relevant states in the presence of strong background electric fields is the same as in their absence, only up to smooth coefficients. In modern bibliography, these states are called Hadamard, (cf Ref. [5]), and they are particularly interesting in the context of quantum field theory on curved spacetimes, where the same idea is used to define observables necessary for the study of e.g. perturbative interacting field theory or semiclassical gravity (cf. [7; 8] for detailed discussions on the definition on such observables).

Semiclassical gravity, a possible first step towards a quantum theory of gravity, couples classical gravity to the expectation value of the stress-energy tensor, which similarly as the vacuum polarization needs to be redefined. To our knowledge, only perturbative corrections to the metric have been studied [13]. In this article we study, as a toy model, the effect of the backreaction of a charged scalar field to a background classical electric field in 1+1 dimensional Minkowski spacetime. This setup was already studied in [1], where it was found

[1] And any other observable defined through the evaluation of products of fields or their derivatives at coinciding points.

that ignoring the backreaction of the scalar field leads to instabilities, and it claims that the inclusion thereof avoids them. However, as pointed out in [11], the expression used in [1] for vacuum polarization ρ cannot be derived from a manifestly gauge invariant prescription. [15] derives the correct expression for ρ, which has qualitative disagreements with the one used in [1].

This discussion motivates the following question, which we answer in the present paper: Given the difference in the prescriptions for ρ, does using the correct one avoid these instabilities? We will use the correct prescription described in [15] to implement an iterative method to study the effect of the backreaction of the scalar field. It has been found that this expression also alleviates the above mentioned instabilities.

The layout of this article is as follows: Section 2 ...

Maybe add?: [2] discusses hadamard states in the pressence of time-like boundaries.

This article summarizes and extends the results of the M.Sc. thesis of Santiago Sanz-Wuhl, written in 2025 at the Institut für Theoretische Physik, Universität Leipzig, under the supervision of Jochen Zahn.

2 Set up

In 1+1 dimensional spacetime, two charges $q, -q$ are set at $x^1 = 0, x^1 = a$, respectively. Ignoring vacuum polarization effects, this yields a constant electric field of strength q pointing towards positive x^1 in the region $x^1 \in [0, a]$, 0 everywhere else. We study the solutions to the semiclassical Klein-Gordon-Maxwell equations (KGM), i.e. the system of coupled differential equations

$$\left(D_\mu D^\mu + m^2\right)\phi = 0, \tag{1a}$$

$$\partial_\mu F^{\mu\nu} = \langle j^\nu \rangle, \tag{1b}$$

in the region $x = (x^0, x^1) \in \mathbb{R} \times [0, a]$. Here, $D_\mu = \partial_\mu + ieA_\mu$ is the gauge covariant derivative,

$$j^\nu(x) = \frac{ie}{2}\left((D^\nu\phi^*(x))\phi(x) - \phi^*(x)D^\nu\phi(x)\right) \tag{2}$$

is the charge current density of the Klein-Gordon field, and $\langle \mathcal{O} \rangle$ is the vacuum expectation value of an observable $\mathcal{O}$, and the metric signature is $(+, -)$. We solve for $F_{\mu\nu}$ and (the mode decomposition of) the two-point function of the vacuum state.

In 1+1 dimensions, A_1 can be arbitrarily set to 0. This way, Eq. (1b) reduces to the Poisson equation for the electrostatic potential A_0

$$\partial_1^2 A_0(x^1) = -\langle j^0 \rangle \equiv -\rho, \tag{3}$$

with ρ the vacuum polarization, i.e. the vacuum expectation value of the zeroth component of the charge current density. Eq. (3) will be solved with the boundary conditions

$$\partial_1 A_0(x^1)\big|_{x^1=0,a} = -q. \tag{4}$$

This way, measuring the electric field close enough to the boundary is not affected by vacuum polarization. The boundary conditions (4) fix the solution to Eq. (3) up to an addition constant, which is for convenience chosen so that $A_0(\frac{a}{2}) = 0$. With these boundary conditions and for a given $\rho(x^1)$, Eq. (3) is solved by

$$A_0(x^1) = -q\left(x^1 - \frac{a}{2}\right) - \int_{\frac{a}{2}}^{x^1}\int_0^{\chi} \rho(\xi)d\xi d\chi. \tag{5}$$

2.1 Field modes

In this gauge, the Klein-Gordon equation is of the form

$$\left[(\partial_0 + ieA_0)^2 + \partial_1^2 - m^2\right]\phi = 0, \tag{6}$$

which we will study using Dirichlet boundary conditions (DBC) $\phi|_{x^1=0,a} = 0$ or Neumman boundary conditions (NBC) $\partial_1\phi|_{x^1=0,a} = 0$. More general boundary conditions may be considered, though they are outside of the scope of this work. Using the mode ansatz $\phi_n(x) = \phi_n(x^1)e^{-i\Omega_n x^0}$ the PDE is split into ordinary differential equations for each mode $\phi_n(x^1)$ of the scalar field

$$\left[(\Omega_n - eA_0)^2 + \partial_1^2 - m^2\right]\phi_n = 0. \tag{7}$$

The following dimensionless parameters are defined

$$z = a^{-1}x^1 \tag{8a}$$

$$\lambda = eEa^2 \tag{8b}$$

$$\varepsilon = ae \tag{8c}$$

$$\omega_n = a\Omega_{n,\alpha} \tag{8d}$$

Eq. (7) then takes the form

$$\left[(\omega_n - \varepsilon A_0)^2 + \frac{d^2}{dz^2} - a^2 m^2\right]\phi_n = 0. \tag{9}$$

The modes ϕ_n are normalized with respect to the symplectic norm

$$(\phi_n, \phi_m) = i\int_0^1 (\phi_n^* \pi_m^* - \pi_n \phi_m) dz, \tag{10}$$

with $\pi_n = (D_0\phi_n)^*$ the canonical conjugate of ϕ_n.

The general solution to Eq. (6) is thus decomposed in modes

$$\phi(t,z) = \sum_{n>0} a_n \phi_n(z) e^{-i\omega_n t} + \sum_{n<0} b_n^\dagger \phi_n(z) e^{-i\omega_n t}. \tag{11}$$

Positive (negative) values of n are identified with modes of positive (negative) norm. a_n and b_n are the ladder operators for the positive and negative norm solutions, respectively, obeying canonical commutation relations $[a_n, a_n^\dagger] = [b_n, b_n^\dagger] = \delta_{nm}$, and all other commutators vanishing. The ladder operators define the vacuum state $|0\rangle$ as that for which $a|0\rangle = b|0\rangle = 0$. Note that due to the chosen gauge, $\omega_n = -\omega_{-n}$, and the $b_n^\dagger$ are the creation operators of the negative frequency family of solutions.

As mentioned above, the definition of quantities such as ρ —Wick powers—is problematic, since it involves the product of local observables evaluated at coinciding spacetime points (see Eq. (2)). To define such polynomials, one assumes that the states of a field ϕ are Hadamard, i.e. they have a similar short-distance behavior to the vacuum of ϕ in absence of background fields, and are assumed to be the only physically relevant states [5]. Under this assumption, the divergence of such states is state independent and locally constructed from the background fields. Since ϕ is coupled to the background field through the covariant derivative, one may use the setup at hand as a toy model for semiclassical gravity, a possible first step towards an elusive quantum theory of gravity.

Specifically, one says that the two point function $w_\Omega^{\phi\phi^*}(x,x') = \langle\Omega|\phi(x)\phi^*(x')|\Omega\rangle$ of the scalar field ϕ in a Hadamard state Ω is of the form

$$w_\Omega^{\phi\phi^*}(x,x') \equiv H^{\phi\phi^*}(x,x') + R_\Omega^{\phi\phi^*}(x,x'). \tag{12}$$

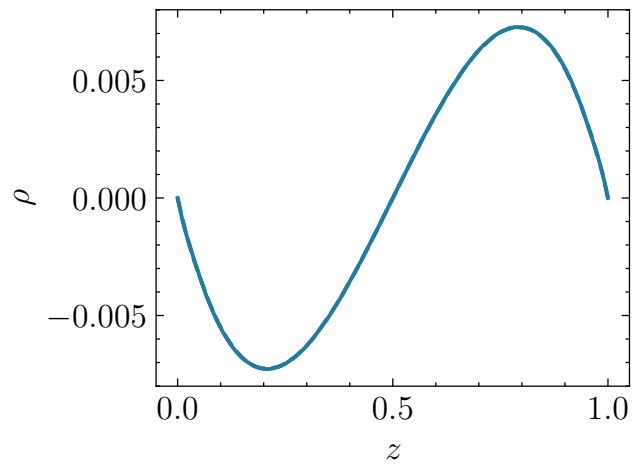

Fig. 1: The vacuum polarization ρ of the scalar field due to a background electrostatic potential $A_0(z) = -0.5(z-0.5)$, with DBC and $m = 0$.

Here,

$$H^{\phi\phi^*} = -\frac{V(x,x')}{4\pi}\ln[(t-t')^2 - (z-z')^2 + i\varepsilon \mathrm{sign}(t-t')] \tag{13}$$

is the Hadamard parametrix of the scalar field in 1+1 spacetime dimensions and $R_\Omega^{\phi\phi^*}$ a state-dependent smooth function. The smooth coefficient V, up to order relevant for vacuum polarization reads [11]

$$V(x,x') = \exp\left[-i\varepsilon\int_0^1 A_\mu(x' + s(x-x'))(x-x')^\mu ds\right] + \mathcal{O}(x-x')^2.$$

The Hadamard condition allows us to define expectation values of local observables non-linear in the fields as a limit of coinciding points:

$$\langle D_\alpha\phi(x) D_\beta^* \phi^*(x)\rangle = \lim_{x'\to x} D_\alpha D_\beta^{*\prime}(w_\Omega^{\phi\phi^*}(x,x') - H^{\phi\phi^*}(x,x')), \tag{14}$$

with α, β symmetrized multiindices.

Eqs. (14), (2) together with the mode decomposition

(11) result in the expression for ρ calculated in [15]

$$\rho = \lim_{\tau\to 0}\sum_{n>0}\Big[(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{-i\omega_n\tau} + \sum_{n<0}(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{i\omega_n\tau}\Big] + \frac{\varepsilon^2}{\pi}A_0. \tag{15}$$

An example of ρ using this expression is given in Fig. 1, calculated in the external field approximation explained below, i.e. a homogeneous background electric field given by the potential $A_0(z) = -0.5(z-0.5)$.

To measure the *strength* of the vacuum polarization, the *induced charge*

$$Q_I = \int_0^{\frac{1}{2}} \rho(z)dz \tag{16}$$

is defined. This charge partially screens the background electric field λ, and Gauß's law yields that the total electric field at $z = \frac{1}{2}$ as $\lambda + Q_I$.

2.2 External Field Approximation and Critical Field Strength

The external field approximation ignores the contribution of the backreaction of the charged scalar field to the background electric field. This approximation displays the instabilities of the system, which the inclusion of backreaction aims to avoid. Eq. (9) is solved with $A_0(z) = -\lambda(z-\frac{1}{2})$, and takes the form

$$\left[\left(\omega_n + \lambda\left(z-\frac{1}{2}\right)\right)^2 + \frac{d^2}{dz^2} - a^2m^2\right]\phi_n = 0. \tag{17}$$

For each value of λ, the modes ϕ_n and their energies ω_n can be obtained from the boundary value problem. Fig. 2 displays the energies $\omega_1, \omega_2, \omega_3$ of the first three modes of the massless scalar field with DBC as functions of the background field strength λ. The value at which ω_1 goes to 0 defines the critical field strength $\lambda_c \approx 16.05039$. Fig. 3 displays the divergent behavior of $Q_I(\lambda)$ for this configuration as $\lambda \to \lambda_c$. λ_c adopts different values for different boundary conditions and masses. [1] shows that for $\lambda > \lambda_c$ some frequencies ω turn complex, which leads to runaway solutions. We will not be concerned by this case.

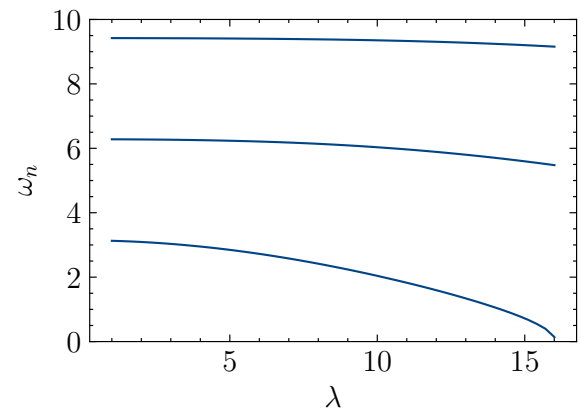

Fig. 2: Energies of the three first modes of the massless Klein-Gordon field subject to DBC in the external field approximation as functions of λ. $\omega_1 \to 0$ as $\lambda \to \lambda_c \approx 16.05$ defines the critical field strength for these mass and boundary conditions configurations.

2.3 Mode sum formula

Vacuum polarization in [1] is defined through a so-called mode sum formula. It pairs the charge density associated to each mode as

$$\rho_n = (\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2. \tag{18}$$

to define the vacuum polarization

$$\rho^{\mathrm{MS}} = \sum_{n>0}\rho_n = \sum_{n>0}\left[(\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2\right], \tag{19}$$

where the superscript MS is introduced to differentiate it from the correct prescription ρ. The above expression cannot be derived from a manifestly gauge invariant renormalization scheme, and it relies on the specific pairing of modes to be well-defined. Furthermore, in the iterative procedure discussed above, [1] truncates the mode sum to $\rho^{\mathrm{MS}} \approx \rho_1$, .

Fig. 4 shows different charge densities arising from the different ρ prescriptions, for the massless

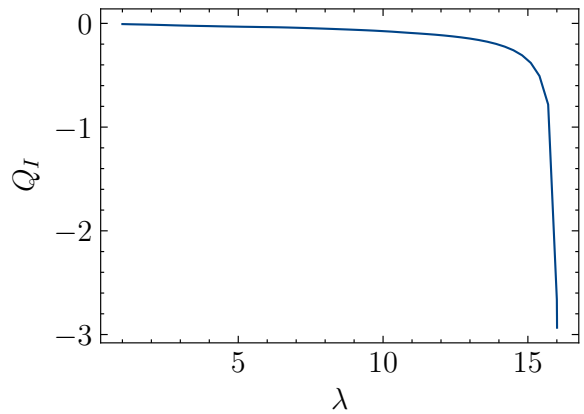

Fig. 3: Corresponding induced charge Q_I in Fig. 2 as a function of λ.

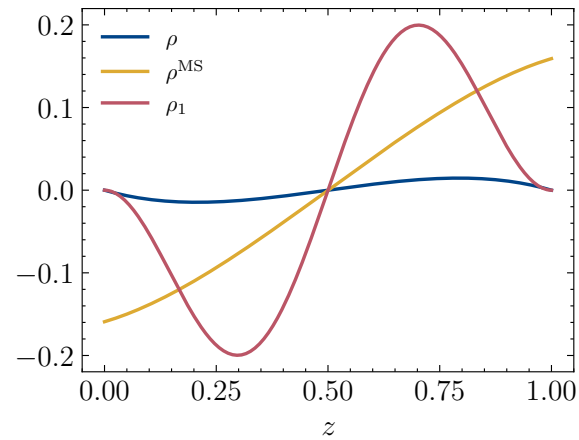

Fig. 4: Three different ρ for the same massless scalar field subject to DBC resulting from $A_0 = -(z-\frac{1}{2})$: In blue, the correct prescription in Eq. (15) used throughout this paper. In orange, ρ calculated through the mode sum (19). In red, ρ defined through the mode sum formula, truncated to the first mode.

scalar field subject to DBC. The same ϕ_n, ω_n and $A_0(z) = -(z-\frac{1}{2})$ have been used for the different ρ. In blue, the vacuum polarization used in the computations throughout this paper, in orange the vacuum polarization arising from the mode pairing and in red the one used for the iterative procedure in [1]. Notice the non-zero vacuum polarization at the boundaries for Eq. (19), which the reader might find surprising to find in a field subject to DBC. This is an artifact of the missing $\frac{\varepsilon^2}{\pi}A_0$ term in Eq. (19).

We observe that ρ is significantly 'weaker' than the one used by [1], where it was seen that the screening due to ρ^{MS} avoids the instabilities appearing in the external field approximation. This motivates the question answered in this paper: Does considering the correct vacuum polarization ρ also avoid these instabilities?

3 Self-consistent fields

As discussed above, the polarization of the vacuum induces a charge density ρ, which modifies the background field as in Eq. (5). This modified background field can be used in Eq. (20a) to calculate new solutions, that will have a different vacuum polarization. This iterative procedure, schematized in Fig. 5, is performed *ad infinitum*, and we study its convergence.

More precisely, for each value of λ and at each iteration κ, we compute the following

$$\frac{d^2\phi^{\kappa}_{n,\lambda}}{dz^2} = -\left[(\omega^{\kappa}_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})^2 - a^2m^2\right]\phi^{\kappa}_{n,\lambda}, \tag{20a}$$

$$\begin{aligned}\rho^{\kappa}_{\lambda} = \lim_{\tau\to 0}\Big[&\sum_{n>0}(\omega^{\kappa}_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^{\kappa}_{n,\lambda}|^2 e^{-i\omega^{\kappa}_{n,\lambda}\tau} \\ &+ \sum_{n<0}(\omega^{\kappa}_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^{\kappa}_{n,\lambda}|^2 e^{i\omega^{\kappa}_{n,\lambda}\tau}\Big] \\ &+ \frac{\varepsilon^2}{\pi}A^{\kappa-1}_{0,\lambda},\end{aligned} \tag{20b}$$

$$\begin{aligned}A^{\kappa}_{0,\lambda} &= cA^{\kappa-1}_{0,\lambda} \\ &\quad + (1-c)\left[-\lambda\left(z-\frac{1}{2}\right) - \int_{\frac{1}{2}}^{z}\int_0^{\chi}\rho^{\kappa}_{\lambda}(\xi)d\xi d\chi\right].\end{aligned} \tag{20c}$$

This procedure has two main differences from the one used in [1]: the expression for the vacuum polarization, and the $c < 1$ prescription in Eq. (20c), which is discussed below.

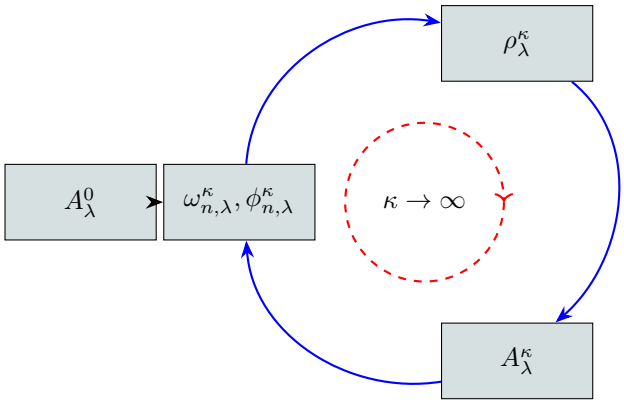

Fig. 5: Schematic of the iterative procedure for fixed λ.

Interpreting Eq. (20c) as an update map on $A_{0,\lambda}^{\kappa-1}$, we identify the iterative scheme with an infinite-dimensional fixed-point problem. That is, for each λ, we seek $A_{0,\lambda}$ such that

$$A_{0,\lambda} = f_\lambda(A_{0,\lambda}), \tag{21}$$

with f_λ given by the composition of steps (20a), (20b) and (20c). The fixed points of f_λ, denominated "self-consistent solutions", are looked for by studying the limit $A_{0,\lambda} = \lim_{\kappa\to\infty} A_{0,\lambda}^\kappa$, where

$$A_{0,\lambda}^{\kappa+1} = f_\lambda(A_{0,\lambda}^\kappa) = f_\lambda^\kappa\left(A_{0,\lambda}^0\right), \text{with } f_\lambda^\kappa = \underbrace{f_\lambda \circ \ldots \circ f_\lambda}_{k \text{ times}}. \tag{22}$$

If the limit $A_{0,\lambda}$ exists, it is given the name self-consistent potential, as well as the corresponding self-consistent vacuum polarization ρ_λ, field modes $\phi_{n,\lambda}$ and mode energies $\omega_{n,\lambda}$.

For low enough $\lambda \ll \lambda_c$, the initial value $A_{0,\lambda}^0$ can be set to the external field approximation $A_{0,\lambda}^0(z) = -\lambda\left(z - \frac{1}{2}\right)$. In the strong-field regime, this is no longer a valid initial condition, due to numerical instabilities appearing for $\lambda > \lambda_c$. This is discussed in the Appendix A. A numerical continuation method is used to traverse the λ-parameter:

1. Assume a self-consistent $A_{0,\lambda}^\infty$ is known.
2. Choose an increment $\Delta\lambda > 0$, which defines a candidate $\lambda_{\text{new}} = \lambda + \Delta\lambda$.
3. Perform the iterative procedure with the initial condition
$$A_{0,\lambda_{\text{new}}}^0 = A_{0,\lambda}^\infty.$$
4. If this sequence converges, store the λ and self-consistent fields, and proceed from step 2. If it fails, reduce $\Delta\lambda$, and proceed from step 1.

Finally, in Eq. (20c), $c \in [0, 1]$ is introduced in order to avoid an overshooting of the potential: For $c = 0$, the procedure is precisely the one used in [1], and it is expected to quickly converge. For c close to 1, updates will be small corrections to the previous potential.

4 Results

Using the iterative procedure above laid out we focus on two particular cases: The massless scalar field subject to DBC, and $m = 1$ with the scalar field subject to NBC. We do not study the massless field subject to NBC, since under these boundary conditions, $\lambda_c = 0$ for $m = 0$, as pointed out in Ref. [15].

Figures 6 and 7 display the dynamics of KGM system in the semiclassical approximation. The external field approximation is given for reference in orange. Notice the different λ scales. The system is studied through the energies of the first modes (top panel), the induced charge Q_I (middle panel) and the total electric field $\lambda + Q_I$ at the midpoint $z = 0.5$ (bottom panel). We observe in both cases how considering backreaction avoids the instabilities reflected in $\omega_1 \to 0$ and $Q_I \to -\infty$ as $\lambda \to \lambda_c$ that the external field approximation presents. Indeed, the energy of the lower modes is raised which "prevents" ω_1 from crossing 0. We also point the reader to notice how the rapid growth of $|Q_I|$ as λ crosses λ_c keeps $\lambda + Q_I$ below λ_c. We find this behavior surprising, as intuitively one would expect $\lambda + Q_I$ to stabilize. Considering a quantum background electric field would perhaps fulfill this intuition.

One might also notice the almost identical behavior of the self-consistent solutions to the external field approximation in the weak-field regime $\lambda \ll \lambda_c$ in both cases.

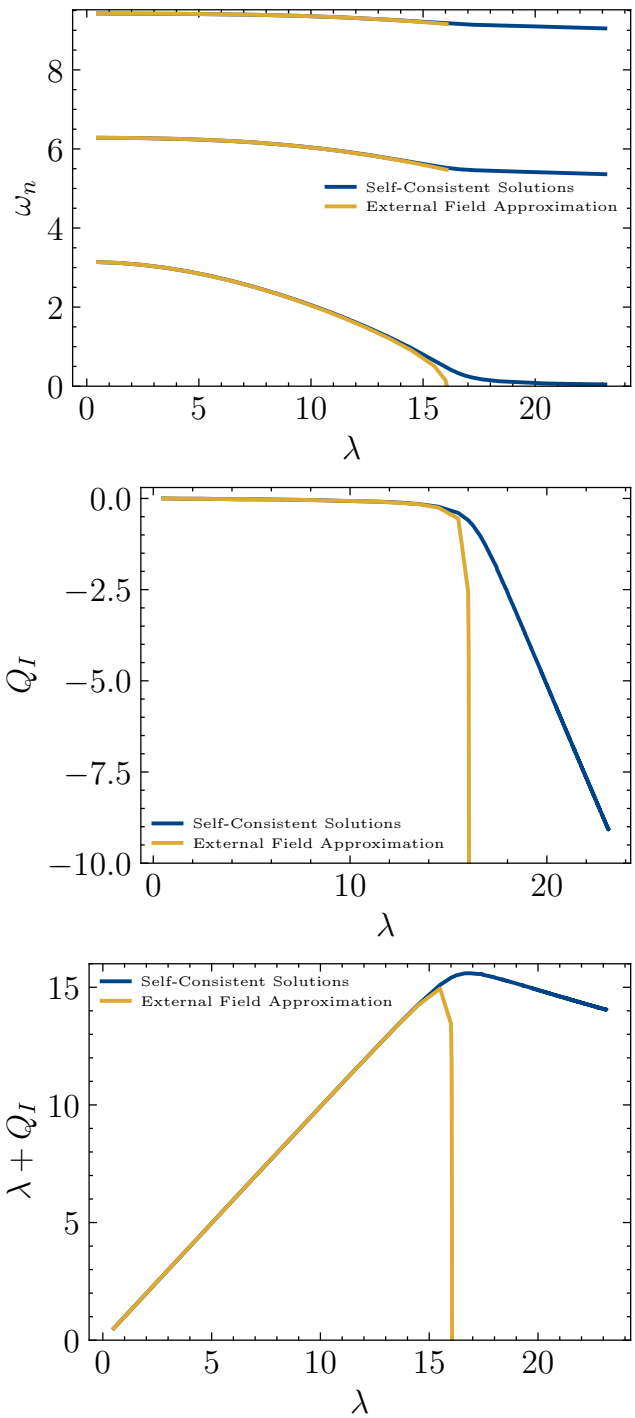

Fig. 6: In blue, the self-consistent solutions to the coupled semiclassical Klein-Gordon-Maxwell system for a massless scalar field subject to DBC. In orange, for reference, the external field approximation is given. Top panel: Energies of the first three modes as functions of λ. Middle panel: Corresponding Q_I for each λ. Bottom panel: Total electric field $\lambda + Q_I$ due to the screening of the vacuum at $z = \frac{1}{2}$.

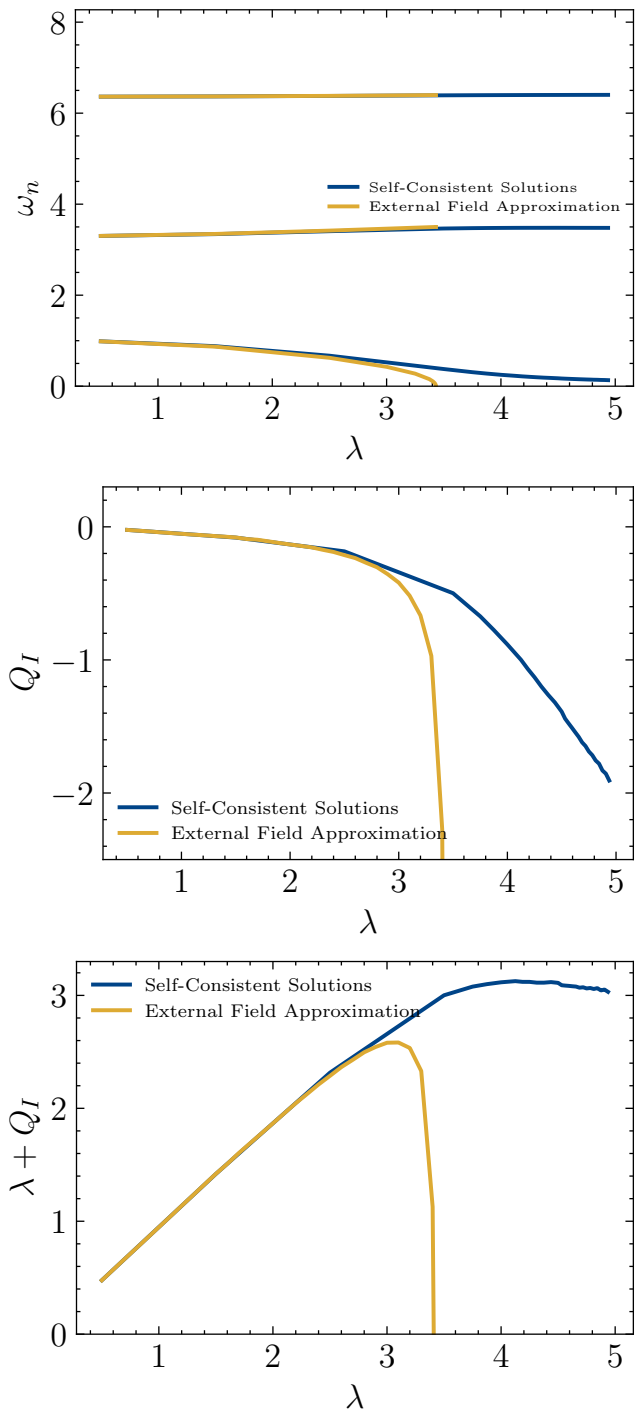

Fig. 7: Following the format in Fig. 6, the results for the unit mass scalar field subject to NBC. Further details in main text.

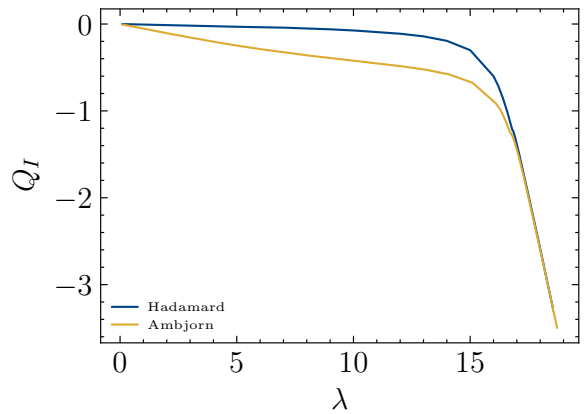

Fig. 8: Q_I for the self-consistent solutions for a massless field subject to DBC, by using different expressions for ρ. In blue, the (correct) vacuum polarization obtained through renormalization through point-splitting w.r.t. a Hadamard parametrix, and in orange the vacuum polarization obtained through a mode sum formula truncated to the first mode used in [1].

4.1 Mode sum formula

For completeness, we compare the results obtained in this work with those of [1]. Fig. 8 displays Q_I in the iterative procedure as a function of λ for two different prescriptions for vacuum polarization: in blue, the correct prescription ρ used throughout this paper, and in orange ρ^1 as discussed in Sec. 2.3, truncated to the first mode. The two prescriptions exhibit markedly different behavior in the weak-field regime, where the truncated mode sum yields a noticeably stronger vacuum polarization. In the strong-field regime, however, their behavior is very similar. This can be understood as a consequence of the low-energy modes being the most sensitive to the background field, and thus the ones that contribute most to vacuum polarization.

5 Discussion and outlook

[1] studied the effect of backreaction of a charged quantum scalar field on a background classical constant electric field in a finite region. It was found that vacuum polarization raises the energy of the quantum field modes, thus avoiding runaway solutions that otherwise appear. It was later pointed out [11] that their prescription for vacuum polarization, relying on a mode sum expansion leads to wrong results. The correct prescription is introduced in [15] which defines vacuum polarization ρ in a manifestly gauge-invariant manner through point-splitting w.r.t. a Hadamard parametrix and finds discrepancies of approximately an order of magnitude between the two prescriptions.

In this paper, we have revisited the semiclassical KGM equations studied by Ambjørn *et al.*, with the correct prescription for ρ. We have systematically explored varying boundary conditions and masses of the scalar field. For this, we have introduced the physical set up, a finite 1-dimensional region with a constant background field generated by two classical charges at the boundaries, and employed a mode ansatz to decompose the field solution ϕ. The correct prescription of ρ in [15] is reviewed. This relies on the assumption that the vacuum state of the scalar field is Hadamard. This allows us to use the Hadamard parametrix, a state-independent, locally constructed object, as the counterterm in the point-splitting renormalization procedure used in defining ρ. Finally, the iterative procedure to solve the KGM equations for a given background field is outlined, and the solution space is traversed by a numerical continuation method.

Our primary finding is that the backreaction through ρ also avoids the instabilities that otherwise appear. However, we observe that beyond the critical field, the total electric field does not stabilize at a fixed value; instead, it recedes. This might indicate a breakdown of the semiclassical approximation, as the quantum fluctuations of the electric field are neglected.

Interestingly, when we compare the results using the correct prescription with those obtained with the older method, their strong-field behaviors are found to be indistinguishable. This is attributed to the low energy modes being the most sensitive to the background field, which are precisely the modes that dominate the onset of runaway solutions.

Future work could focus on benchmarking the semiclassical approximation with scalar QED. This is a key comparison delineating the regime of validity of the semiclassical approach.

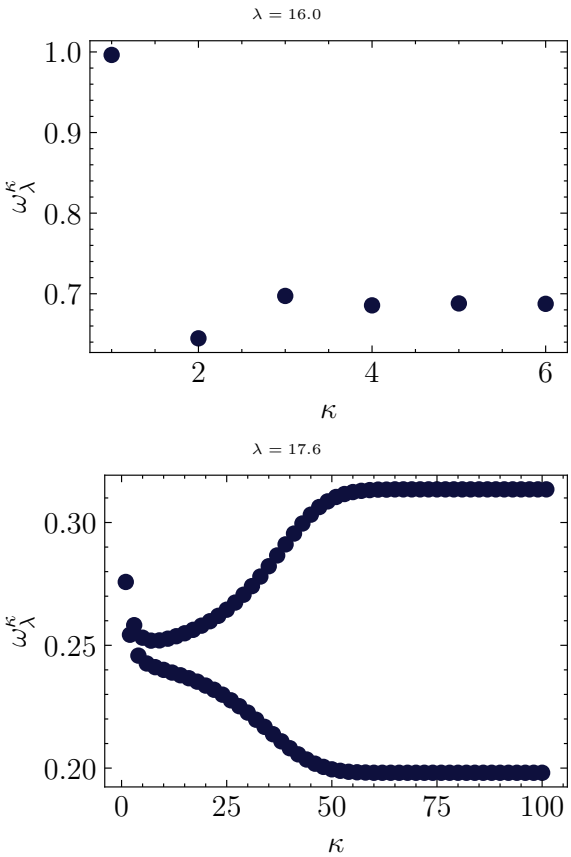

Fig. 9: ω_1 of the massless scalar field with DBC for different values of λ as a function of the iteration κ.

A Convergence of the self-consistent fields

The iterative procedure either converges, oscillates, or the appearance of complex ω cuts the sequence short.

A case where the sequence convergences, and oscillates is displayed in Fig. 9 in the top, and bottom panel, respectively. We measure convergence in the frequency of the lowest mode ω_λ^κ. These convergence cases can be understood from the perspective of fixed-point theory. For brevity, we invite the reader to find the fixed points of the $\mathbb{R} \to \mathbb{R}$ functions $\frac{3}{4} - \frac{x}{2}, 1 - x$ to see how the slope of the function affects convergence. The parameter c in Eq. (20c) allows us to control this slope in order to ensure convergence.

Complex ω might appear if the candidate $\lambda + \Delta\lambda$ in the numerical continuation method is too far away from λ. The screening due to ρ_λ is "too weak" compared to $A^1_{0,\lambda+\Delta\lambda}$, which makes the system effectively behave as in the external field approximation with $\lambda > \lambda_c$.

References

[1] J. Ambjorn and S. Wolfram. Properties of the Vacuum. 2. Electrodynamic. *Annals Phys.*, 147: 33, 1983. doi: 10.1016/0003-4916(83)90066-0.

[2] B. Costeri, C. Dappiaggi, B. A. Juárez-Aubry, and R. D. Singh. The Hadamard parametrix on half-Minkowski with Robin boundary conditions: Fundamental solutions and Hadamard states, Sept. 2025. URL `http://arxiv.org/abs/2509.26035`. arXiv:2509.26035 [math-ph].

[3] P. A. M. Dirac. Discussion of the infinite distribution of electrons in the theory of the positron. *Mathematical Proceedings of the Cambridge Philosophical Society*, 30(2):150–163, 1934. doi: 10.1017/S030500410001656X.

[4] A. Fedotov, A. Ilderton, F. Karbstein, B. King, D. Seipt, H. Taya, and G. Torgrimsson. Advances in QED with intense background fields. *Physics Reports*, 1010:1–138, Apr. 2023. ISSN 03701573. doi: 10.1016/j.physrep.2023.01.003. URL `http://arxiv.org/abs/2203.00019`. arXiv:2203.00019 [hep-ph].

[5] C. J. Fewster and R. Verch. The Necessity of the Hadamard Condition. *Class. Quant. Grav.*, 30:235027, 2013. doi: 10.1088/0264-9381/30/23/235027. _eprint: 1307.5242.

[6] W. Heisenberg and H. Euler. Consequences of Dirac's theory of positrons. *Z. Phys.*, 98(11-12): 714–732, 1936. doi: 10.1007/BF01343663. _eprint: physics/0605038.

[7] S. Hollands and R. M. Wald. Local Wick Polynomials and Time Ordered Products of Quantum Fields in Curved Spacetime. *Communications in Mathematical Physics*, 223(2):289–326,

Oct. 2001. ISSN 0010-3616, 1432-0916. doi: 10.1007/s002200100540. URL http://arxiv.org/abs/gr-qc/0103074. arXiv:gr-qc/0103074.

[8] S. Hollands and R. M. Wald. Quantum fields in curved spacetime. *Phys. Rept.*, 574:1–35, 2015. doi: 10.1016/j.physrep.2015.02.001. _eprint: 1401.2026.

[9] W. E. Lamb and R. C. Retherford. Fine Structure of the Hydrogen Atom by a Microwave Method. *Phys. Rev.*, 72:241–243, 1947. doi: 10.1103/PhysRev.72.241.

[10] P. J. Mohr, G. Plunien, and G. Soff. QED corrections in heavy atoms. *Phys. Rept.*, 293(5-6):227–369, 1998. doi: 10.1016/s0370-1573(97)00046-x.

[11] J. Schlemmer and J. Zahn. The current density in quantum electrodynamics in external potentials. *Annals Phys.*, 359:31–45, 2015. doi: 10.1016/j.aop.2015.04.006. _eprint: 1501.05912.

[12] J. S. Schwinger. On gauge invariance and vacuum polarization. *Phys. Rev.*, 82:664–679, 1951. doi: 10.1103/PhysRev.82.664.

[13] J. C. Thompson and E. Winstanley. Quantum-corrected anti–de Sitter spacetime. *Physical Review D*, 110(12):125003, Dec. 2024. ISSN 2470-0010, 2470-0029. doi: 10.1103/PhysRevD.110.125003. URL https://link.aps.org/doi/10.1103/PhysRevD.110.125003.

[14] E. A. Uehling. Polarization effects in the positron theory. *Phys. Rev.*, 48:55–63, 1935. doi: 10.1103/PhysRev.48.55.

[15] J. Wernersson and J. Zahn. Vacuum polarization near boundaries. *Phys. Rev. D*, 103(1):016012, 2021. doi: 10.1103/PhysRevD.103.016012. _eprint: 2010.05499.